

L9b: Pricing American Style Options Contracts using the Cox, Ross, Rubinstein (CRR) Binomial Model

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced European options, which can be exercised only at expiration. Standard options on stocks and ETFs are American-style and can be exercised at any time. What is the right to exercise early worth?

Today's concepts

- Distinguish between European and American options: exercise only at expiration, or at any time before expiration.
- Apply the binomial model to price American options: build CRR trees and compute premiums with the decision rule for early exercise.
- Estimate the premium of American options by simulation: Longstaff–Schwartz Monte Carlo, regressing the value of holding on the share price.

Lectures and Examples

Lecture:
CHEME-5660-L9b-Lecture-American-Derivatives-Intrinsic-Extrinsic-Value-Fall-2026.ipynb

Example 1:
CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Example 2:
CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Review on your own:
CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Company Profile: The Options Clearing Corporation

OCC stands between the buyer and the seller of every U.S. listed option. It opened in 1973 as CBOE Clearing Corporation and became OCC in 1975.

- Exercise and assignment: OCC assigns each early exercise at random to a clearing member with a short position in the same contract.
- Automatic exercise at expiration: since 1977, an expiring equity option in the money by 0.01 USD/share or more is exercised unless the holder instructs otherwise.
- Monte Carlo margin and binomial trees: since 2006, STANS simulates 10,000 two-day scenarios and reprices American options on a binomial tree.

Connection to today's lecture: a seller of an American put can be assigned once exercising now beats holding, so a European-premium margin would leave out the early-exercise premium.

American Style Options Contracts

American-style options contracts allow the holder to exercise the option at any time before or on the expiration date. Previously, we wrote the premium of an American-style contract as:

$$\mathcal{P}^{\text{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\Big) \geq \mathcal{P}^{\text{EU}}$$

The exercise time τ is chosen using only the information available at that time (a stopping time). The American premium is at least the European one because the American holder can always wait until expiration.

The standard method to price American options is the Cox, Ross, Rubinstein (CRR) binomial model.

Reference: Cox, J. C., Ross, S. A., and Rubinstein, M. (1979). Option pricing: A simplified approach. Journal of Financial Economics 7(3): 229.

CRR Binomial Lattice Model

At each time step, the share price moves up by a factor u with the risk-neutral probability q, or down by a factor d with probability $(1-q)$. The CRR parameters are no longer data driven: q makes the expected return of the underlying asset match the risk-free rate g_y.

$$
\begin{aligned}
\mathcal{D}_{1,0}(g_y) \cdot S_0 &= \mathbb{E}_{\mathbb{Q}}\left(S_1\right) && \text{grow at the risk-free rate under } \mathbb{Q} \\
&= q \cdot (u \cdot S_0) + (1-q) \cdot (d \cdot S_0) && \text{up with probability } q\text{, down with } (1-q) \\
&= S_0(qu + (1-q)d) && \text{factor out } S_0 \\
&= S_0(qu + d - qd) && \text{expand } (1-q)d \\
&= S_0(d + q(u-d)) && \text{collect the } q \text{ terms} \\
\implies & \boxed{q = \frac{\mathcal{D}_{1,0}\left(g_y\right) - d}{u-d}} && \text{divide by } S_0 \text{ and solve for } q
\end{aligned}
$$

Here $\mathcal{D}_{1,0}(g_y) = e^{g_y \Delta t}$ is the risk-free accumulation factor for one time step. This is the risk-neutral probability we found by ruling out arbitrage when we introduced lattice models, with $R_f = \mathcal{D}_{1,0}(g_y)$.

CRR Up and Down Factors

Unlike the data-driven binomial model we discussed earlier, the CRR model derives u and d based on the volatility σ of the underlying asset, the time step Δt, and the symmetric condition $ud = 1$:

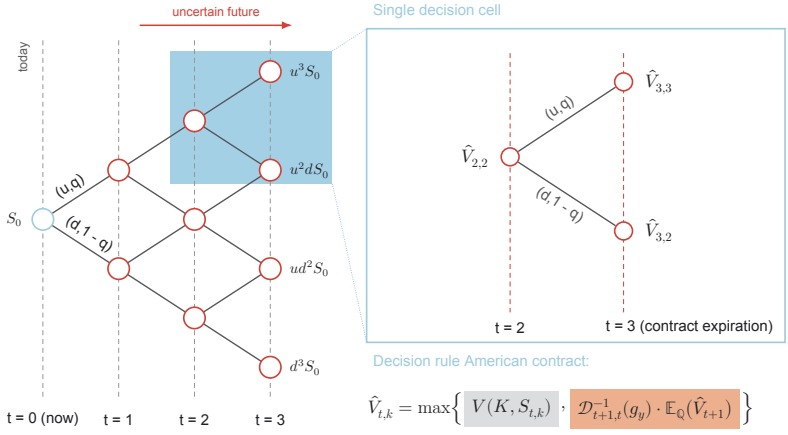
$$\boxed{\begin{aligned} u &= e^{\sigma\sqrt{\Delta t}} \\ d &= e^{-\sigma\sqrt{\Delta t}} \end{aligned}}$$

Here σ is the same volatility as in geometric Brownian motion and the Black–Scholes–Merton formula (units: $\mathrm{yr}^{-1/2}$).

Where do u and d come from? advanced/crr_factors/CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Decision-Rule for American Options

At each node, the American holder decides to exercise now or hold for the future.



The Early-Exercise Premium

Working backward from the payoffs at expiration, the American holder takes the larger branch, while the European holder can only hold:

$$\hat{V}^{\mathsf{AM}}_{t,k} = \max\Big\{ \underbrace{V(K, S_{t,k})}_{\text{exercise now}},\ \underbrace{\mathcal{D}^{-1}_{t+1,t}(g_y) \cdot \overbrace{\big(q\,\hat{V}^{\mathsf{AM}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{AM}}_{t+1,k}\big)}^{\mathbb{E}_{\mathbb{Q}}}}_{\text{hold for the future}} \Big\}$$

$$\hat{V}^{\mathsf{EU}}_{t,k} = \mathcal{D}^{-1}_{t+1,t}(g_y) \cdot \overbrace{\big(q\,\hat{V}^{\mathsf{EU}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{EU}}_{t+1,k}\big)}^{\mathbb{E}_{\mathbb{Q}}}$$

Here $\hat{V}_{t,k}$ is the contract's value with k up moves in t steps, $S_{t,k} = S_0 u^k d^{t-k}$, and $\mathcal{D}^{-1}_{t+1,t}(g_y) = e^{-g_y \Delta t}$. The premium is $\mathcal{P} = \hat{V}_{0,0}$, and $\mathcal{P}^{\mathsf{AM}} - \mathcal{P}^{\mathsf{EU}} \geq 0$ is the early-exercise premium.

Pseudocode: Pricing on the CRR Lattice

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, and the number of time steps h. Compute $\Delta t = T/h$, u and d, q, and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

1. Build the share price lattice: $S_{t,k} \leftarrow S_0 u^k d^{t-k}$ for each level $t = 0, 1, \ldots, h$ and node $k = 0, 1, \ldots, t$.
2. At expiration, set each node value to the payoff: $\hat{V}_{h,k} \leftarrow V(K, S_{h,k})$ for $k = 0, 1, \ldots, h$.
3. For each level $t = h-1, h-2, \ldots, 0$ **do**:
   - For each node $k = 0, 1, \ldots, t$ **do**:
     a. Compute the value of holding for the future: $H_{t,k} \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot (q\,\hat{V}_{t+1,k+1} + (1-q)\,\hat{V}_{t+1,k})$.
     b. If the contract is American, **then**: $\hat{V}_{t,k} \leftarrow \max\{V(K, S_{t,k}),\ H_{t,k}\}$. Otherwise, $\hat{V}_{t,k} \leftarrow H_{t,k}$.
4. The premium is the value at the root: $\mathcal{P} \leftarrow \hat{V}_{0,0}$.

A Hypothetical Example from Hull

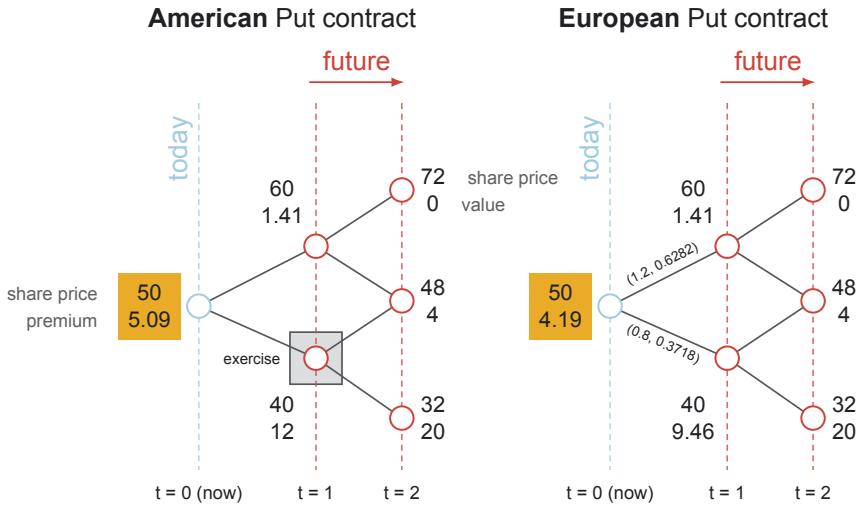
Consider a put contract.

- Contract: the strike price is $K = 52$ USD/share on a stock whose current price is $S_0 = 50$ USD/share.
- Lattice: two time steps of 1 year each. In each period, the stock price moves up by a factor of $u = 1.2$ or down by a factor of $d = 0.8$.
- Pricing inputs: the continuously compounded risk-free rate is $g_y = 5\%$, so the risk-neutral probability of an up move is $q = (e^{0.05} - 0.8)/(1.2 - 0.8) = 0.6282$.

What is the price of the American and European put option contracts?

Source: Chapter 13 of Hull, Options, Futures, and Other Derivatives, 9th edition (2015).

American versus European Put on One Lattice



Early Exercise in the Hull Example

At the down node at $t = 1$, exercising pays 12 USD/share, while holding is worth 9.46 USD/share, so the American holder exercises.

That one decision raises the premium from 4.19 to 5.09 USD/share, an early-exercise premium of about 0.90 USD/share.

Let's compute every node of this lattice, then apply the same method to contracts from the course's options archive.

Review on your own: CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

We'll work through in lecture: CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Monte Carlo Pricing of American Options

We simulate many share price paths under $\mathbb{Q}$, then estimate the expected value of holding at each date by regressing the discounted cash flows from holding on the current share price across paths. Like the lattice, Longstaff and Schwartz work backward from expiration:

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, the number of exercise dates h after today, and the number of paths M. Compute $\Delta t = T/h$ and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$ (expiration), where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths, fit $\hat{C}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{C}_t(S_i(t))$, **then**: exercise, so $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let’s price an American put by simulation and compare it with the lattice and the European premium.

Holding on every path gives the European Monte Carlo premium, so the difference estimates the early-exercise premium.

Example 2: CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Optional Advanced Material

CRR factors: advanced/crr_factors/
CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

In this notebook, we derive the up and down factors of the CRR model by matching the mean and variance of the lattice's one-step growth rate to those of geometric Brownian motion. We then show that the lattice still matches geometric Brownian motion when we price with the risk-neutral probability, and find how small the time step must be to keep that probability between zero and one.

Notebooks: lectures/week-9/L9b/advanced, also linked from the lecture notebook.

Summary

Today we priced American options with the CRR binomial model and Monte Carlo simulation.

Key takeaways

- Binomial trees price American options: at each node, the holder exercises or holds, so backward induction computes the premium.
- The decision rule sets the early-exercise premium: in Hull's put, exercising at the one node where it beats holding accounts for the whole gap between the American and European premiums.
- Simulation estimates the same premium: Longstaff–Schwartz works backward along simulated paths, regressing the value of holding on the share price. Holding on every path gives the European premium.

The early-exercise premium is what the right to exercise before expiration is worth, and the lattice and simulation give two ways to compute it.